Math 50

4.1 Study design & confounding 4.2 Multiple-predictor regression 4.3 Covariance matrix & estimation
4.4 Sample distribution & collinearity 4.5 Categorical predictors Problems

Unit 4

Study design, multiple predictors, covariance matrices, collinearity, and categorical data

This unit covers linear regression with **multiple predictors**. We will first discuss the idea of a **confounding variable** and study design, which motivates models with multiple predictors. Many new ideas emerge when we add more predictors. Most importantly, regression coefficients now need to be interpreted as average differences with all *other* predictors held fixed, so the relationship between predictors plays a big role. After this unit you should understand how adding and removing predictors changes the others. We'll talk about the joint sample distribution of regression coefficients and **collinearity**, understanding these ideas from multiple angles (linear algebra, geometry, and simulation). We'll also introduce **categorical predictors**, which let us handle qualitative data. We will also discuss the linear algebra view of fitting a regression model.

CONCEPTS

Study design, randomized controlled trials and observational studies, confounding, regression with multiple predictors, controlling for a variable, the covariance matrix and the OLS estimator, Simpson's paradox, the effect of adding predictors on R^2, the joint sample distribution of the coefficients, collinearity, categorical predictors and dummy variables.

THINGS TO PRACTICE

- Derive the linear system relating the regression coefficients to the covariances (generalizing the single-predictor case).
- Interpret the output of a fitted model with multiple predictors, including qualitative predictors.
- Predict how adding a predictor changes the regression coefficient of an existing predictor, given some knowledge of the predictors.

- Identify collinearity and understand its effect on the sample distribution.
- Include categorical predictors in a regression model.

Now we're ready to study **linear regression with multiple predictors**. Much of the theory carries over from the single-predictor case, but there are new subtleties in how we *interpret* the predictors. In particular, a regression coefficient represents an average difference in the response variable when all other predictors are held fixed: this is the idea of **controlling for** another variable. We'll also work out what happens to the model when we add or remove predictors.

4.1 Study design and confounding

A central question in many scientific contexts is: Does changing variable X cause a change in variable Y?

It should be clear that two variables can be strongly related in a dataset without either one causing the other. The classic textbook example is the relationship between ice cream sales and drowning. They are associated, but tempurature is a **confounding variable** (or confounder). Whether we are able to interpret a relationship as causal depends less on the regression formula itself than on how the data were collected, or what we call the **study design**.

Here, to give a taste of this topic, we will discuss only two types of study designs:

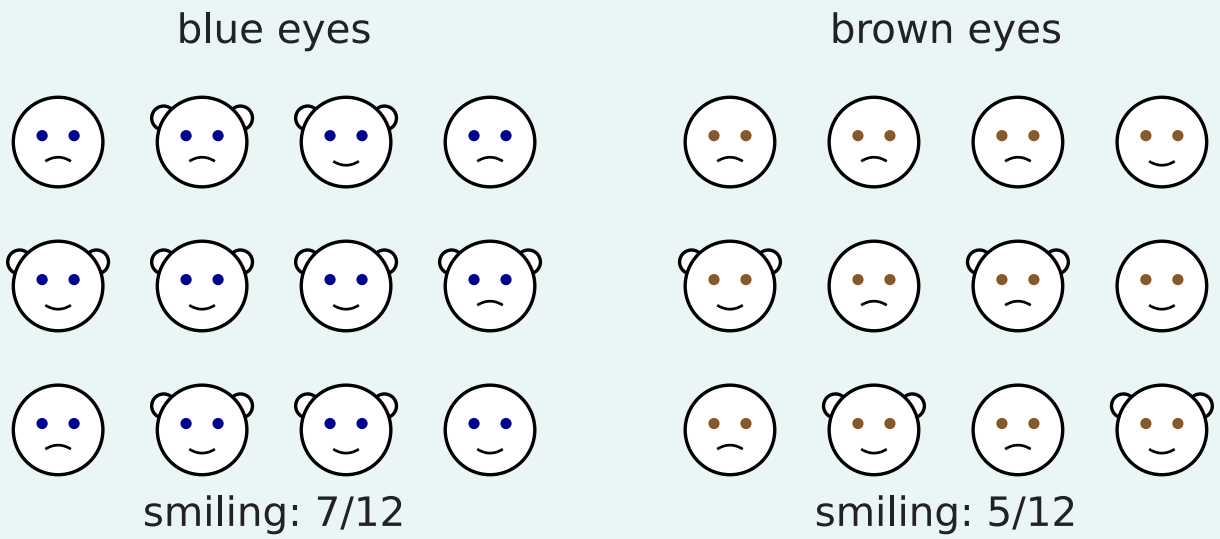
Randomized controlled trial (RCT). If we randomly assign each unit's value of X — e.g. assigning patients to treatment or control — then X is, by construction, independent of other variables that might affect Y. Any systematic difference in Y between the groups can then be attributed to X itself.

Cohort study (prospective observational). Often randomization is not possible or ethical. A cohort study follows a group forward in time and observes who ends up with which value of X and what Y they go on to have. Because X was observed rather than assigned, it may be related to other causes of Y.

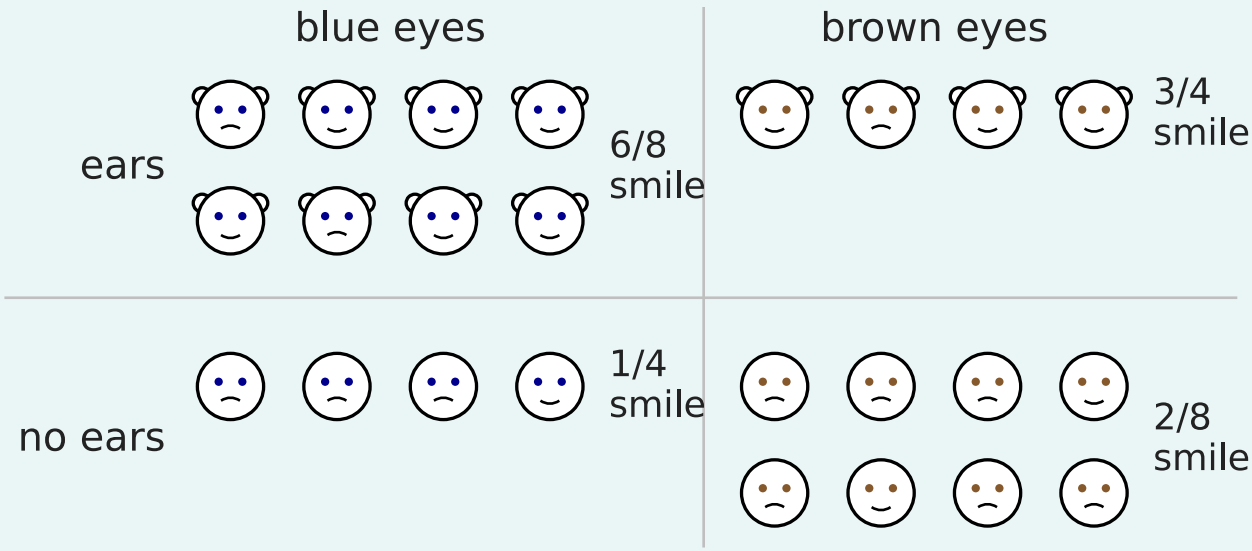
In a cohort study we cannot know whether there is another variable confounder X_2, correlated with X, which also has an effect on Y. It can produce an association between X and Y even if X has no causal effect on Y, or it can inflate, shrink, or reverse a real effect. In an RCT, randomization makes X independent of *every* such X_2, measured or not. (We call it X_2 rather than a new letter because the fix will be to include it as a second predictor in the regression.)

Example

Recall the population of people from Unit 1, who differ in eye color and whether they smile. Let $X = 1$ if a person has blue eyes and $Y = 1$ if they smile:



Blue-eyed people smile more often: $P(Y = 1 \mid X = 1) = 7/12$ but $P(Y = 1 \mid X = 0) = 5/12$. Do blue eyes make people smile? Now also record $X_2 = 1$ if a person has ears and $X_2 = 0$ if they do not:



Among people with the *same* ears, eye color makes no difference: $3/4$ smile with ears and $1/4$ without, whatever their eyes. Blue-eyed people are simply more likely to have ears ($8/12$ vs. $4/12$), and ears go with smiling. So X_2 is a confounder: it produces an association between eye color and smiling even though, among people with the same ears, there is none. In regression form (see the next section), the conditional averages are exactly

$$E[Y \mid X] = \frac{5}{12} + \frac{1}{6}X, \qquad E[Y \mid X, X_2] = \frac{1}{4} + 0 \cdot X + \frac{1}{2}X_2,$$

so once we control for ears, the coefficient of eye color drops from $1/6$ to 0.

Aside: other study designs (optional)

There are many other study designs: A **case-control study** works backward from the outcome by sampling people who already have $Y = 1$ (cases) and $Y = 0$ (controls), then looks back at their history of X. A **cross-sectional study** measures X and Y at a single point in time, with no notion of before and after. A **natural experiment** looks for a real-world situation where X was effectively randomized by circumstance, such as a lottery, policy threshold, or eligibility rule.

Drill

Identifying study designs

For each study, name the design and say whether a difference in average Y between the X groups can be read causally.

a. Patients are assigned to a drug or a placebo by a coin flip, and blood pressure is measured six weeks later.

b. A group of adults is recruited, their coffee drinking is recorded, and they are followed for twenty years to see who develops heart disease.

c. Everyone in a town is surveyed on one day about both their exercise habits and their current weight.

Confounding variables

A dataset shows that towns with more ice-cream shops per capita have more drownings per capita.

a. Propose a variable X_2 that plausibly affects both, and explain why it fits the definition of a confounder.

b. Would randomly assigning the number of ice-cream shops remove the problem? Explain in one sentence.

c. Give one variable that is affected by X and Y but is *not* a confounder.

4.2 Multiple-predictor regression

The real power of regression comes from models of the form

$$Y = \beta_0 + \sum_{i=1}^{K} \beta_i X_i + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma^2),$$

where X_i is one of K predictor variables. Equivalently,

$$Y \mid (X_1 = x_1, \ldots, X_K = x_K) \sim \text{Normal}\left(\beta_0 + \sum_{i=1}^{K} \beta_i X_i,\ \sigma^2\right).$$

You might see the shorthand $Y \sim \text{LR}(X, \beta, \sigma^2)$.

In these notes, our goal is to answer:

1. What are the estimators of the parameters in this model?
2. How do we interpret the **regression coefficients** β_i?
3. What is the sample distribution of the regression coefficients?
4. How do correlations between predictors influence the answers to these questions?

Example (two-predictor regression in `statsmodels`)

Question: generate data from the two-predictor model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma^2),$$

and fit the linear regression $Y \sim X_1 + X_2$ using `statsmodels`.

Solution:

```
import numpy as np
import statsmodels.api as sm

# --- simulate data ---
n = 300
rng = np.random.default_rng(0)

beta0, beta1, beta2, sigma = 1.0, 2.0, -1.0, 0.5
X1 = rng.normal(size=n)
X2 = rng.normal(size=n)
eps = rng.normal(scale=sigma, size=n)
Y = beta0 + beta1*X1 + beta2*X2 + eps

# --- fit OLS: Y ~ X1 + X2 ---
X = sm.add_constant(np.column_stack([X1, X2]))
model = sm.OLS(Y, X)
results = model.fit()
```

Recall from Section 3.4 that $R^2 = 1 - \mathrm{SSR}/\mathrm{SST}$, where $\mathrm{SSR} = \sum_i r_i^2$ is the sum of the squared residuals r_i (the differences between each Y_i and its fitted value) and $\mathrm{SST} = \sum_i (Y_i - \overline{Y})^2$ is the total sum of squares. R^2 is the fraction of the variation in Y explained by the model. If $R^2 = 1$ the fitted values match the data exactly, and if $R^2 = 0$ the model does no better than predicting $\overline{Y}$ for every data point.

The interpretation of R^2 is the same as before, except now we're comparing the variance conditioned on *all* predictors to the overall variance in Y:

$$R^2 = 1 - \frac{\sum_i r_i^2}{\sum_i (Y_i - \overline{Y})^2} \approx 1 - \frac{\mathrm{var}(Y \mid X_1, X_2)}{\mathrm{var}(Y)},$$

where, in the multi-predictor case, the residuals are

$$r_i = Y_i - \left(\hat{\beta}_0 + \sum_{k=1}^{K} \hat{\beta}_k X_{i,k}\right).$$

Basic interpretation and estimation of the parameters

To interpret the parameters, it's easiest to work with just two predictors, as in the example above. The conditional expectation of Y is

$$E[Y \mid X] = \beta_0 + \beta_1 X_1 + \beta_2 X_2,$$

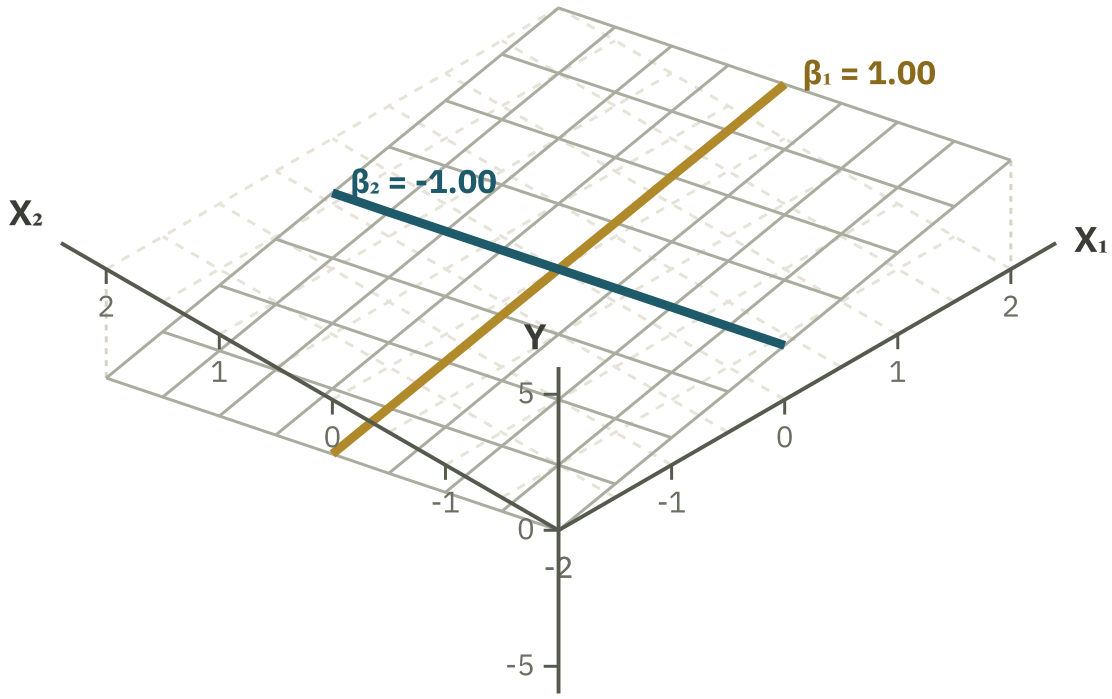
where I'm using the shorthand $E[Y \mid X] = E[Y \mid (X_1, X_2)]$ to mean the expectation conditioned on **both** predictors. This is the equation for a flat surface (a plane) over the (x_1, x_2)-plane. If we slice the surface in the x_1 direction and look at it from the side, we see a line with slope β_1 (and similarly for x_2). This gives the key interpretation:

$$\beta_1 \text{ is the slope of } E[Y \mid X] \text{ as } X_1 \text{ varies and } X_2 \text{ is held fixed.}$$

The demo below slices the surface both ways: the orange line holds X_2 fixed and varies X_1, the blue line holds X_1 fixed and varies X_2. Drag either slider and watch its own line's *slope* stay put while only its height moves — that invariance is exactly what the statement above says.

Notice that even though we're conditioning on both variables, the slope β_1 doesn't depend on which value of X_2 we condition on — we get the interpretation of β_2 by swapping the roles of X_1, X_2. This is a consequence of linearity, and it's one of the core assumptions of multiple-predictor regression that we didn't encounter with a single predictor: the "effect" of X_1 doesn't depend on the value of X_2 (and vice versa).

The regression surface $E[Y \mid X] = \beta_0 + \beta_1 X_1 + \beta_2 X_2$



Example (children's test scores)

We'll now work with a new example: children's test scores. Imagine we're studying what factors determine children's success in school, in order to design interventions that help struggling students. The predictors are mother's IQ and whether the mother attended high school. Here, the model assumption says that the association between the mother's high school education and test scores isn't influenced by the mother's IQ: if we compare two random children whose mothers have the same IQ but differ in whether they attended high school, the average *difference* in test scores won't depend on the IQ of their mothers — although the average test score itself will.

Question: fit the data to a linear regression model with two predictors and answer:

a. What are the regression coefficients, and their interpretations?

b. Based on this analysis, which factor — IQ or high school education — seems more predictive of test scores?

c. Overall, how well do high school education and IQ together predict test scores?

d. What's the chance a student whose mother has an IQ of 90 and did not attend high school does better than a student whose mother has an IQ of 110 and did attend high school?

Solution: we get the following output from `statsmodels` (see the companion Colab notebook):

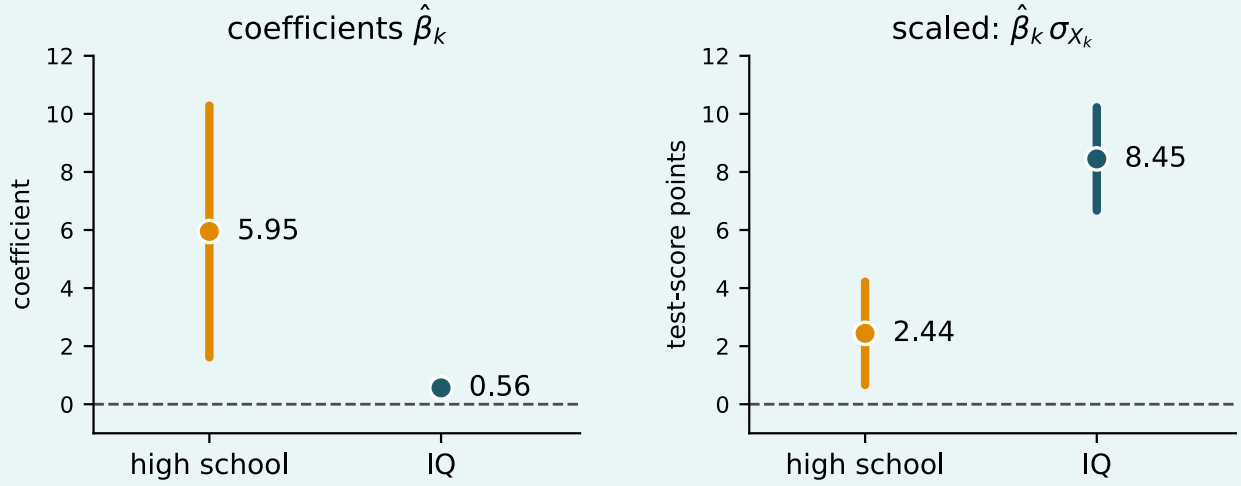
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.
------------------------------------------------------------------------------
const         25.7315      5.875      4.380      0.000      14.184      37
mom_hs         5.9501      2.212      2.690      0.007       1.603      10
mom_iq         0.5639      0.061      9.309      0.000       0.445       0
```

a. For the coefficients:

- $\beta_{\text{hs}} \approx 5.95$: among students whose mothers **have the same IQ**, a student whose mother attended high school scores 5.95 points higher on average than one whose mother didn't.
- $\beta_{\text{iq}} \approx 0.56$: among students whose mothers **have the same high-school education status**, a one-point difference in mother's IQ is associated with a 0.56-point difference in score, on average.
- $\hat{\beta}_0 \approx 26$: mathematically, the average score of students whose mother didn't attend high school and has zero IQ — not a meaningful quantity, since nobody has zero IQ, so we can ignore it when interpreting the output.

b. β_{hs} is about ten times larger, but the two coefficients have different units: X_{iq} ranges roughly from 70 to 130, while X_{hs} is 0 or 1. To compare them, multiply

each coefficient by the standard deviation of its predictor, $\hat{\beta}\hat{\sigma}_X$. This is the standard deviation of the term βX across the population: the typical amount by which that predictor moves a randomly chosen student's expected score away from the average, with the other predictor held fixed. Here $\hat{\sigma}_{\text{hs}} \approx 0.41$ and $\hat{\sigma}_{\text{iq}} \approx 15$, so $\hat{\beta}_{\text{hs}}\hat{\sigma}_{\text{hs}} \approx 5.95 \times 0.41 \approx 2.44$ vs. $\hat{\beta}_{\text{iq}}\hat{\sigma}_{\text{iq}} \approx 0.56 \times 15 \approx 8.45$: the association with IQ is stronger.



For a binary predictor, $\hat{\sigma}_X$ depends on how common each value is: a rare binary predictor has a small $\hat{\sigma}_X$ even if its effect on the few people who have it is large. Other conventions exist; for example, Gelman (2008) multiplies continuous predictors by $2\hat{\sigma}_X$, to match a $0/1$ predictor with $\hat{\sigma}_X \approx 0.5$. Here that gives 5.95 for high school vs. $0.56 \times 30 \approx 16.9$ for IQ, so IQ still matters more.

c. The R^2 value is 0.214, so about 21% of the variation in test scores is explained by mother's high-school education and IQ together.

d. In the Colab notebook, we calculate this probability to be about 25%.

This interpretation generalizes to many predictors. In terms of conditional expectations,

$$\beta_i = E[Y \mid X_1, \ldots, X_{i-1}, X_i{=}x_i{+}1, X_{i+1}, \ldots, X_K] - E[Y \mid X_1, \ldots, X_{i-1}, X_i{=}x_i, X_{i+1},$$

a natural extension of the two-predictor formula below.

Relationship between single- and two-predictor regression coefficients

We can write the two-predictor coefficient explicitly in terms of conditional averages,

$$\beta_1 = E[Y \mid X_1{=}(x{+}1), X_2] - E[Y \mid X_1{=}x, X_2].$$

How is this related to covariance? A first guess: just as in the single-predictor case, $\beta_1 = \text{cov}(Y, X_1)/\sigma_{X_1}^2$. After all, slicing the 2D planar surface along x_1 gives the same slope β_1 for every x_2, so it stands to reason that regressing on the (x_1, y) points alone should also give slope β_1. **But this argument silently assumes that when x_1 changes, x_2 doesn't change with it.** Let's see why that matters with an example.

Example (test scores: multiple vs. single predictor)

We again consider children's test scores, comparing the two-predictor fit above to a fit using only high-school education as a predictor.

Question: what's the difference between the coefficient of X_{hs} when it's the only predictor and when X_{iq} is also included? How are the two related?

Solution: using only mother's high-school education, we get $\hat{\beta}'_{\text{hs}} \approx 12$ and $\hat{\beta}'_0 \approx 78$ (using β' for the single-predictor coefficients), so $\hat{y} = 12X_{\text{hs}} + 78$ — while with both predictors, the coefficient of X_{hs} is about half that.

In the one-predictor model, a coefficient of 12 means a student whose mother went to high school does 12 points better on average, i.e. we're predicting

$$E[Y \mid X_{\text{hs}}{=}1] - E[Y \mid X_{\text{hs}}{=}0] = \beta'_{\text{hs}} \approx 12.$$

Compare this to the two-predictor model. The average test score for students whose mothers went to high school is

$$\begin{aligned}\hat{y}_{\text{hs}} &\approx E[Y \mid X_{\text{hs}}{=}1] \\ &= E[\beta_0 + \beta_{\text{hs}} + \beta_{\text{iq}}X_{\text{iq}} \mid X_{\text{hs}}{=}1] \\ &= \beta_0 + \beta_{\text{hs}} + \beta_{\text{iq}}E[X_{\text{iq}} \mid X_{\text{hs}}{=}1] \\ &\approx 26 + 6 + 0.6\,\overline{X}_{\text{iq|hs}}.\end{aligned}$$

where $\overline{X}_{\text{iq}|\text{hs}}$ is the sample average IQ among mothers who attended high school. Similarly, $\hat{y}_{\text{no-hs}} = 26 + 0.6\,\overline{X}_{\text{iq}|\text{no-hs}}$. So, according to the two-predictor model, the average difference between the groups is

$$\Delta\hat{y}_{\text{hs}} = 6 + 0.6\big(\overline{X}_{\text{iq}|\text{hs}} - \overline{X}_{\text{iq}|\text{no-hs}}\big),$$

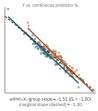
or, in probabilistic notation,

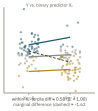
$$E[Y \mid X_{\text{hs}}{=}1] - E[Y \mid X_{\text{hs}}{=}0] = \beta_{\text{hs}} + \beta_{\text{iq}}\big(E[X_{\text{iq}} \mid X_{\text{hs}}{=}1] - E[X_{\text{iq}} \mid X_{\text{hs}}{=}0]\big).$$

We can compute $\overline{X}_{\text{iq}|\text{hs}} - \overline{X}_{\text{iq}|\text{no-hs}} \approx 10.3$, giving $\Delta\hat{y}_{\text{hs}} \approx 12$ — recovering the single-predictor coefficient from the two-predictor model.

The demo below separates the two slopes, using a binary X_1 and continuous X_2 (as in the KidIQ example above). The left panel plots Y against X_2, split by X_1: the two solid lines are the within-group slopes, which should both sit close to β_2. The right panel plots Y against X_1, split by tercile of X_2: the solid segments are the within-tercile differences, which should sit close to β_1. In both panels, the dashed line is the marginal slope from regressing Y on that one predictor alone — increase the confounding strength b to pull the solid and dashed lines apart.

Marginal slope vs. the regression coefficient





$X_1 \sim \text{Bernoulli}(0.5)$, $X_2 \mid X_1 \sim \text{Normal}(bX_1, 1)$, $Y \mid (X_1, X_2) \sim \text{Normal}(\beta_1 X_1 + \beta_2 X_2,\ 0.4^2)$. Left: Y vs. the continuous predictor X_2, split by X_1. Right: Y vs. the binary predictor X_1 (jittered), split by tercile of X_2. Increase b to strengthen the confounding between the two predictors and pull the solid (within-group) and dashed (marginal) lines apart.

The important thing above is that the two predictors are *not* independent. If they were, $\overline{X}_{\text{iq}|\text{hs}} - \overline{X}_{\text{iq}|\text{no-hs}}$ would be zero, and the coefficient of X_{hs} would have to be the same with or without X_{iq} in the model. This generalizes: for any model where X_1 is a binary predictor,

$$\beta_1' = \beta_1 + \beta_2\big(E[X_2 \mid X_1{=}1] - E[X_2 \mid X_1{=}0]\big),$$

where β_1' is the regression coefficient of X_1 *without* X_2 in the model.

Simpson's paradox

In the example above, β_1' and β_1 can even have different signs, depending on the relationship between the predictors. This effect is called **Simpson's "paradox"** — not really a paradox, just a consequence of correlation between predictors shifting the single-predictor coefficient, as illustrated above.

Example (Simpson's paradox with two binary predictors)

Consider two binary predictors $X_1, X_2 \in \{0, 1\}$ with joint distribution

$P(X_1, X_2)$	$X_2 = 0$	$X_2 = 1$
$X_1 = 0$	0.4	0.1
$X_1 = 1$	0.1	0.4

and suppose $Y \mid (X_1, X_2) = X_1 - 2X_2 + \epsilon$.

Question: compute the single-predictor regression coefficient β_1' of X_1 on Y.

Solution: first, the conditional means of X_2 given X_1:

$$P(X_2{=}1 \mid X_1{=}1) = \frac{0.4}{0.1+0.4} = 0.8, \qquad P(X_2{=}1 \mid X_1{=}0) = \frac{0.1}{0.4+0.1} = 0.2,$$

$$\beta_{1,2} \equiv E[X_2 \mid X_1{=}1] - E[X_2 \mid X_1{=}0] = 0.8 - 0.2 = 0.6.$$

With $\beta_1 = 1, \beta_2 = -2$, the single-predictor slope is

$$\beta_1' = \beta_1 + \beta_2\beta_{1,2} = 1 + (-2)(0.6) = -0.2.$$

Note the sign flip: the two-predictor model says increasing X_1 *increases* Y, but the single-predictor model says the opposite!

Effect of adding predictors on R^2

Adding a new predictor to a regression model can never *decrease* R^2. Let $R^2_{1\,\text{pred}}$ be the R^2 for $Y = \beta_1 X_1 + \epsilon'$, and $R^2_{2\,\text{pred}}$ the R^2 for $Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon$. Since $R^2 = 1 - \text{var(residual)}/\,\text{var}(Y)$, R^2 increases whenever the residual variance decreases.

The key point: the error term ϵ' in the reduced (single-predictor) model is *not* the same as $\beta_2 X_2 + \epsilon$ — it absorbs the variation in Y that's correlated with X_2 but unaccounted for once we drop it. Since X_1, X_2 are generally correlated, part of the systematic variation explained by X_2 gets treated as noise in the single-predictor model:

$$\text{var}(\epsilon') = \text{var}(Y \mid X_1) = \text{var}(\beta_1 X_1 + \beta_2 X_2 + \epsilon \mid X_1) = \beta_2^2\ \text{var}(X_2 \mid X_1) + \sigma_\epsilon^2 \geq \sigma_\epsilon^2,$$

hence $R^2_{2\,\text{pred}} \geq R^2_{1\,\text{pred}}$, with equality only when the extra term vanishes: either $\beta_2 = 0$ (X_2 has no effect on Y) or $\text{var}(X_2 \mid X_1) = 0$ (X_2 is already determined by X_1). So adding predictors either improves the fit or leaves it unchanged — never worse.

Drill

Reading a two-predictor fit

A regression of monthly rent Y (in dollars) on floor area X_1 (in square feet) and distance from campus X_2 (in miles) gives $\hat{\beta}_0 = 400$, $\hat{\beta}_1 = 1.2$, $\hat{\beta}_2 = -90$, and $R^2 = 0.62$.

a. State the interpretation of $\hat{\beta}_1$ in one sentence, being explicit about what is held fixed.

b. Predict the rent of a 700 sq ft apartment 1.5 miles from campus.

c. What does $R^2 = 0.62$ say, and what does it *not* say?

Dropping a predictor

For a binary predictor X_1, the two-predictor fit gives $\beta_1 = 2$ and $\beta_2 = 5$, and the predictors satisfy $E[X_2 \mid X_1{=}1] - E[X_2 \mid X_1{=}0] = -0.4$.

a. Compute the single-predictor coefficient β_1' of X_1 alone.

b. Someone who only fits X_1 concludes that X_1 is unrelated to Y. In one sentence, say what they have missed.

c. What would β_1' be if X_1 and X_2 were independent?

4.3 Covariance matrix and estimation of regression coefficients

Here we derive formulas for the regression coefficients in terms of the covariances between predictors and between predictors and the response. This (1) clarifies the relationship between single- and multi-predictor coefficients, and (2) gives us an estimator for the multi-predictor case. We're generalizing the single-predictor relationship $\text{cov}(X, Y) = \beta_1 \sigma_X^2$.

Consider a two-predictor model, and set $\beta_0 = E[X_1] = E[X_2] = 0$ for simplicity (these cancel out in the end — check for yourself that everything still works when they're not

zero!). Then $\text{cov}(X_1, Y) = E[X_1 Y]$, and just as in the single-predictor case (but now conditioning on *both* predictors in the tower property),

$$\begin{aligned}\text{cov}(X_1, Y) &= E[X_1 Y] \\ &= E[X_1 E[Y \mid X_1, X_2]] \\ &= E[X_1(\beta_1 X_1 + \beta_2 X_2)] \\ &= \beta_1 E[X_1^2] + \beta_2 E[X_1 X_2] \\ &= \beta_1 \sigma_{X_1}^2 + \beta_2 \,\text{cov}(X_1, X_2).\end{aligned}$$

Doing the same for X_2 gives two equations:

$$\text{cov}(X_1, Y) = \beta_1 \sigma_{X_1}^2 + \beta_2 \,\text{cov}(X_1, X_2), \qquad \text{cov}(X_2, Y) = \beta_2 \sigma_{X_2}^2 + \beta_1 \,\text{cov}(X_1, X_2).$$

As in the single-predictor case, it's useful to write β_1, β_2 as expectations we can estimate by averaging over data. These two equations can be rewritten with the **covariance matrix**

$$\Sigma = \begin{bmatrix} \sigma_{X_1}^2 & \text{cov}(X_1, X_2) \\ \text{cov}(X_1, X_2) & \sigma_{X_2}^2 \end{bmatrix},$$

as

$$\begin{bmatrix} \text{cov}(X_1, Y) \\ \text{cov}(X_2, Y) \end{bmatrix} = \Sigma \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}.$$

This is consistent with the formula above: dividing the first equation by $\text{var}(X_1)$ recovers $\beta_1' = \beta_1 + \beta_2 \beta_{1,2}$, where $\beta_{1,2} = \text{cov}(X_1, X_2)/\text{var}(X_1)$ is the (single-predictor) regression coefficient of X_1 with X_2 as the response. This generalizes to many predictors: in general, Σ is a $K \times K$ matrix with entries $\Sigma_{i,j} = \text{cov}(X_i, X_j)$.

Letting Σ^{-1} denote the matrix inverse ($\Sigma^{-1}\Sigma = I$), we can solve for the coefficients:

$$\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \Sigma^{-1} \begin{bmatrix} \text{cov}(X_1, Y) \\ \text{cov}(X_2, Y) \end{bmatrix} = \frac{1}{\sigma_{X_1}^2 \sigma_{X_2}^2 - \text{cov}(X_1, X_2)^2} \begin{bmatrix} \sigma_{X_2}^2 & -\text{cov}(X_1, X_2) \\ -\text{cov}(X_1, X_2) & \sigma_{X_1}^2 \end{bmatrix}$$

Compare this to the single-predictor formula $\beta_1' = \text{cov}(X, Y)/\text{var}(X)$: here Σ^{-1} plays the role of $1/\text{var}(X)$, and the vector of βs and Y-X covariances play the roles of β_1 and

$\mathrm{cov}(X, Y)$. If $\Sigma = I$ (uncorrelated, unit-variance predictors), each β_i can be solved for separately, recovering the single-predictor formulas.

We don't usually bother with closed-form solutions once there are many predictors — they get complicated fast. But for two predictors,

$$\beta_1 = \frac{\mathrm{cov}(X_1, Y)\sigma^2_{X_2} - \mathrm{cov}(X_2, Y)\,\mathrm{cov}(X_1, X_2)}{\sigma^2_{X_1}\sigma^2_{X_2} - \mathrm{cov}(X_1, X_2)^2},$$

which, if all variances are set to one, becomes

$$\beta_1 = \frac{1}{1 - \rho^2_{1,2}}\big(\rho_1 - \rho_{1,2}\rho_2\big),$$

where ρ_i is the correlation coefficient between X_i and Y, and $\rho_{1,2}$ is the correlation coefficient between X_1, X_2. If X_1, X_2 are uncorrelated ($\rho_{1,2} = 0$), this reduces to the usual single-predictor connection.

Example (correlated predictors)

Consider the model

$$X_1 \sim \mathrm{Normal}(0, 1), \qquad X_2 \mid X_1 \sim \mathrm{Normal}(bX_1,\, 1 - b^2),$$

$$Y \mid (X_1, X_2) \sim \mathrm{Normal}(\beta_1 X_1 + \beta_2 X_2,\, \sigma^2),$$

with $b \in [0, 1]$. Note $b = \beta_{1,2}$: the single-predictor regression coefficient of X_1 with X_2 as the response, which here (both predictors having unit variance) is also the correlation coefficient.

Question:

a. Show that $\mathrm{var}(X_1) = \mathrm{var}(X_2) = 1$ and $\mathrm{cov}(X_1, X_2) = b$.

b. Express β_1 in terms of b, $\mathrm{cov}(X_1, Y)$, and $\mathrm{cov}(X_2, Y)$.

c. Find the single-predictor coefficient β_1' when only X_1 is used.

Solution:

a. By definition $\mathrm{var}(X_1) = 1$, and $\mathrm{var}(X_2) = b^2\mathrm{var}(X_1) + (1 - b^2) = 1$; also $\mathrm{cov}(X_1, X_2) = b\,\mathrm{var}(X_1) = b$.

b. Substituting into the formula above, $\beta_1 = \dfrac{\mathrm{cov}(X_1, Y) - \mathrm{cov}(X_2, Y)b}{1 - b^2}$.

c. The single-predictor slope of Y on X_1 is $\beta_1 + b\beta_2$.

Estimating regression coefficients

Now suppose we have data $\{(X_{i,1}, \ldots, X_{i,K}, Y_i)\}_{i=1}^N$. Form the **design matrix** with samples in rows and predictors in columns,

$$X = \begin{bmatrix} X_{1,1} & \cdots & X_{1,K} \\ \vdots & \ddots & \vdots \\ X_{N,1} & \cdots & X_{N,K} \end{bmatrix} \in \mathbb{R}^{N \times K}, \qquad Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_N \end{bmatrix} \in \mathbb{R}^N.$$

Assuming X, Y are centered (no intercept), the empirical covariance quantities are

$$\hat{\Sigma} = \frac{1}{N} X^\top X \in \mathbb{R}^{K \times K}, \qquad \hat{c} = \frac{1}{N} X^\top Y \in \mathbb{R}^K.$$

When $\hat{\Sigma}$ is invertible, this gives the closed-form **ordinary least squares** (OLS) estimator,

$$\hat{\beta} = \hat{\Sigma}^{-1}\hat{c} = (X^\top X)^{-1} X^\top Y.$$

Aside: the Moore–Penrose pseudoinverse (optional)

If $\hat{\Sigma}$ is singular (e.g. under exact collinearity, or if $K > N$), we instead use the Moore–Penrose pseudoinverse,

$$\hat{\beta} = X^+ Y, \qquad X^+ = V\Sigma^+ U^\top \quad \text{for } X = U\Sigma V^\top \text{ (the SVD of } X),$$

where Σ^+ inverts the nonzero singular values and leaves the zero ones unchanged.

Drill

Solving the covariance system

Suppose $\sigma^2_{X_1} = \sigma^2_{X_2} = 1$, $\text{cov}(X_1, X_2) = 0.5$, $\text{cov}(X_1, Y) = 1$, and $\text{cov}(X_2, Y) = 0.5$.

a. Write out Σ and solve $\Sigma\beta = (\text{cov}(X_1, Y), \text{cov}(X_2, Y))^\top$ for β_1, β_2.

b. Compute the single-predictor coefficient $\beta_1' = \text{cov}(X_1, Y)/\sigma^2_{X_1}$.

c. Here $\beta_1' = \beta_1$ even though the predictors are correlated. Which term in $\beta_1' = \beta_1 + \beta_2\beta_{1,2}$ explains that?

Reading a statsmodels fit

Consider the following `statsmodels` fit:

```
import statsmodels.api as sm
X_design = sm.add_constant(X)
fit = sm.OLS(Y, X_design).fit()
beta = fit.params
```

a. What column does `sm.add_constant(X)` add, and which regression coefficient does it represent?

b. What does `fit.params` contain?

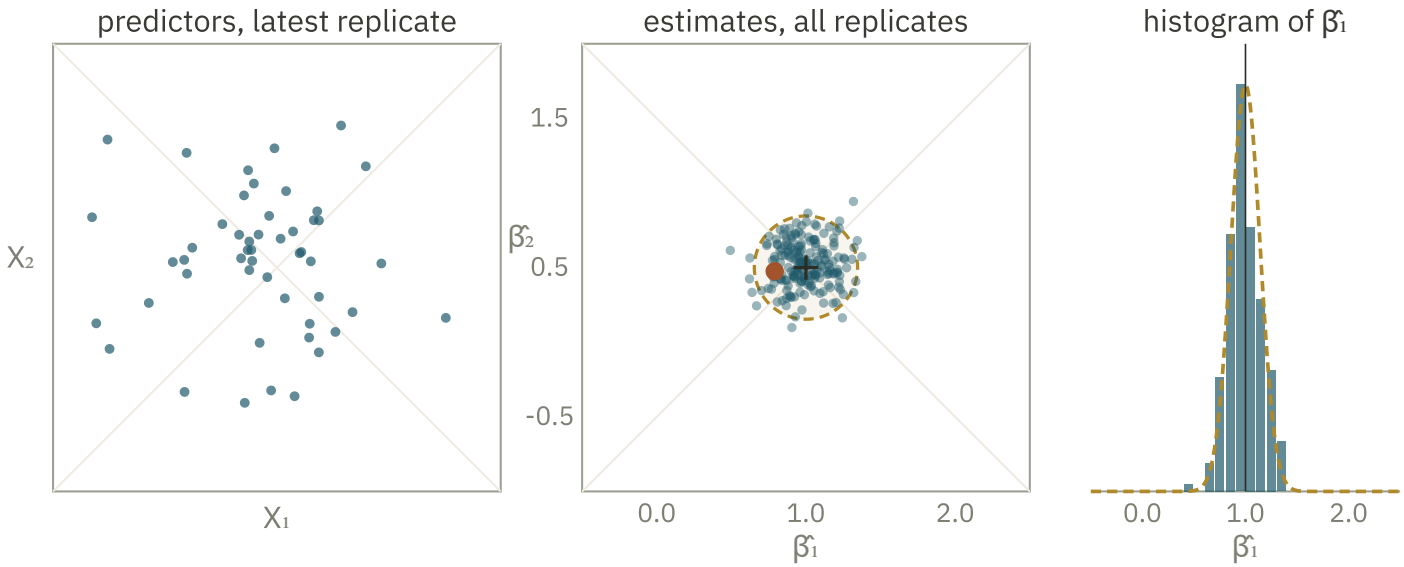
c. Use `fit.summary()` to identify the estimated coefficient, standard error, and p-value for each predictor.

4.4 Sample distribution and collinearity

Just as before, we want to understand the sample distribution of the coefficients — but now we need the *joint* distribution of $(\hat\beta_1, \hat\beta_2, \ldots, \hat\beta_K)$. We'll focus on the two-predictor case. In the demo below, each replicate is a fresh dataset of N points from $Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon$ with $\beta_1 = 1$, $\beta_2 = 0.5$, $\sigma_\epsilon = 1$, and standardized predictors with

the chosen correlation. The left panel shows the predictors of the latest replicate; the middle panel shows $(\hat{\beta}_1, \hat{\beta}_2)$ for every replicate so far (the latest in red), on fixed axes, with the 95% region predicted by the formula below (dashed); the right panel is the histogram of $\hat{\beta}_1$ alone. Start at correlation 0 and press Play: the cloud is round. Then raise the correlation toward 1: the predictors line up along $X_1 = X_2$, and the cloud stretches along the line $\hat{\beta}_1 + \hat{\beta}_2 = \text{const}$, so $\hat{\beta}_1$ on its own becomes much less precise even though the sum $\hat{\beta}_1 + \hat{\beta}_2$ stays precise. A negative correlation stretches the cloud the other way, so the difference is precise and the sum is not. The readouts compare the simulated standard errors with the formula.

The joint sample distribution of $(\hat{\beta}_1, \hat{\beta}_2)$



To build intuition, imagine X_1, X_2 are very highly correlated (if perfectly correlated, we say they're **collinear**). Then

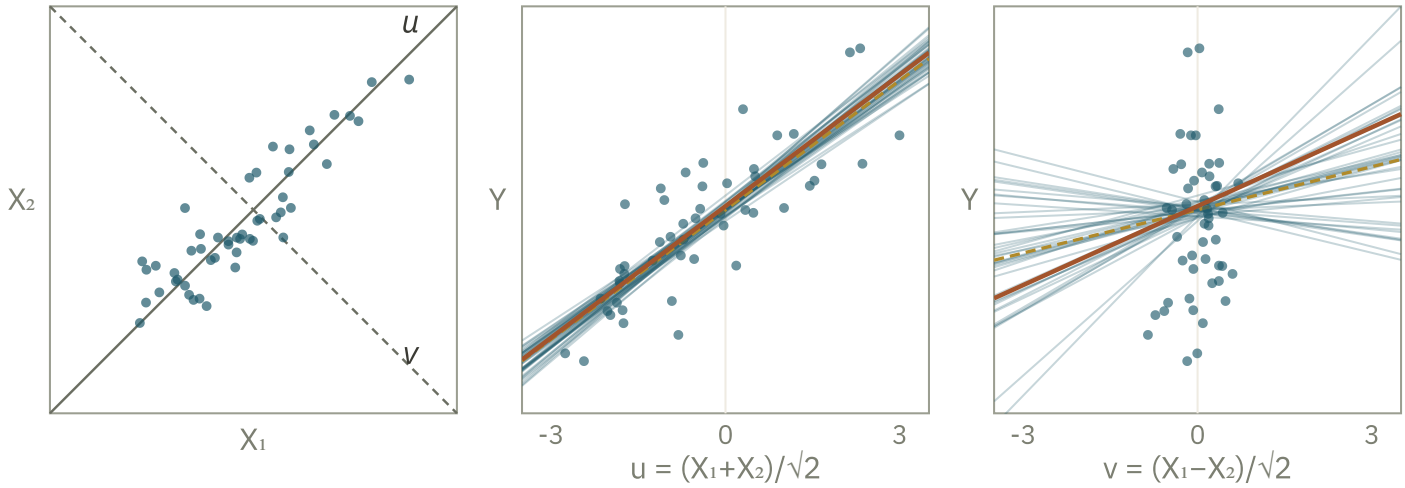
$$Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon \approx \beta_1 X_1 + \beta_2 X_1 + \epsilon = (\beta_1 + \beta_2) X_1 + \epsilon.$$

There are many ways to choose β_1, β_2 so the surface $\beta_1 x_1 + \beta_2 x_2$ stays close to the data, since a change in β_1 can be compensated by an opposite change in β_2. **This means that if**

we estimate $\hat{\beta}_1, \hat{\beta}_2$ and then regenerate new data from the same model (in particular, with the same correlation between X_1 and X_2), we could get very different values of $\hat{\beta}_1, \hat{\beta}_2$, so long as $\hat{\beta}_1 + \hat{\beta}_2$ stays close to what we got before.

To see why, look at the data along two new directions: $u = (X_1 + X_2)/\sqrt{2}$, the direction in which the predictors move together, and $v = (X_1 - X_2)/\sqrt{2}$, the direction in which they disagree. In the demo below, the left panel shows the predictors with these two directions drawn in; the other two panels plot Y against u and against v on the same scale, with the fitted slope for the latest dataset (red), for earlier datasets (faint), and the true slope (dashed). With a high correlation the points are spread out along u but bunched up near $v = 0$, so the slope along u is pinned down and the slope along v is not.

The data seen along u and v



To see why the second view fans out, rewrite the model in terms of u and v:

$$\beta_1 X_1 + \beta_2 X_2 = \frac{\beta_1 + \beta_2}{\sqrt{2}}\, u + \frac{\beta_1 - \beta_2}{\sqrt{2}}\, v,$$

so the slope of the surface in the u view is $(\beta_1 + \beta_2)/\sqrt{2}$ and in the v view it is $(\beta_1 - \beta_2)/\sqrt{2}$. If X_1, X_2 have unit variance and correlation b, then u and v are

uncorrelated with

$$\text{var}(u) = 1 + b, \qquad \text{var}(v) = 1 - b.$$

Recall that in a single-predictor regression the standard error of the slope is $\sigma_\epsilon/(\sigma_X\sqrt{N})$, so a predictor with a small variance gives a slope with a large standard error. The same holds for u and v separately. When b is close to 1, the data points lie near the line $X_1 = X_2$. They are spread out along u, so the slope in the u view has a small standard error. They have very little spread along v, so many different slopes in the v view fit the data about equally well, and the fitted lines fan out.

When b is negative, the points lie near the line $X_1 = -X_2$. Now they are spread out along v and have little spread along u, so the first view fans out and the second stays tight. In terms of the coefficients, for b near -1 the difference $\hat{\beta}_1 - \hat{\beta}_2$ is well determined and the sum $\hat{\beta}_1 + \hat{\beta}_2$ is not.

Sample distribution formula

It can be shown that, assuming the covariance matrix Σ is known, $\hat{\beta}$ has a multivariate Normal distribution:

$$\hat{\beta} \sim \text{Normal}_K\left(\beta,\ \frac{\sigma_\epsilon^2}{N}\Sigma^{-1}\right).$$

If $\Sigma = I$, this reduces to the sample distribution we found in the single-predictor case.

Drill

Correlation and standard errors

Take two standardized predictors with correlation ρ, so

$$\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

Use $\hat{\beta} \sim \text{Normal}_2\left(\beta,\ (\sigma_\epsilon^2/N)\Sigma^{-1}\right)$ with $\sigma_\epsilon^2 = 1$ and $N = 100$.

a. Write $\text{se}(\hat{\beta}_1)$ as a function of ρ.

b. Evaluate it at $\rho = 0$ and at $\rho = 0.9$.

c. What is the sign of the correlation between $\hat{\beta}_1$ and $\hat{\beta}_2$ when $\rho > 0$, and what does that mean about how the two estimates move together across replicates?

Which view fans out?

Two standardized predictors have correlation ρ. As in the demo "Two views of the same fitted planes", let $u = (X_1 + X_2)/\sqrt{2}$ and $v = (X_1 - X_2)/\sqrt{2}$.

a. Compute $\text{var}(u)$ and $\text{var}(v)$ when $\rho = 0.9$. Along which direction are the data more spread out?

b. For $\rho = 0.9$, which is better determined by the data: $\hat{\beta}_1 + \hat{\beta}_2$ or $\hat{\beta}_1 - \hat{\beta}_2$? Which view fans out?

c. Repeat (a) and (b) for $\rho = -0.9$.

d. For which value of ρ do the two views look the same?

4.5 Categorical predictors

Multiple-predictor models often arise when predicting Y from a categorical predictor, like race. In this case, we need to convert the categories to numerical values. If there are two categories (e.g. YES/NO), we map them to 0 or 1. With 3 categories (e.g. White, Black, Other), we might think to map them to $0, 1, 2$ — but this has a problem: a change from 1 to 2 shouldn't necessarily be "the same size" as a change from 0 to 1. **There's no natural ordering of the x values.** We call such predictors **qualitative** rather than **quantitative**, since they express a quality of the data point rather than a numerical quantity.

To handle this, we create **dummy variables**: a set of indicator variables, one per category (minus one, as we'll see). In Python, `pd.get_dummies` does this conversion.

Example (racial disparities in earnings)

We'll fit the earnings data to a model with race as a predictor: what's the association between race and earnings among US adults? We could use a single binary predictor (e.g. White vs. non-White), but that's limiting. Instead, we make one dummy variable per race category. The dataset has 4 categories, $\{\text{Black}, \text{White}, \text{Hispanic}, \text{Other}\}$. In principle we could create a binary variable for each:

$$Y = \beta_0 + \beta_{\text{black}} X_{\text{black}} + \beta_{\text{hispanic}} X_{\text{hispanic}} + \beta_{\text{other}} X_{\text{other}} + \beta_{\text{white}} X_{\text{white}} + \epsilon,$$

but this is problematic: at least one of these predictors must equal 1 for every observation, so they're perfectly collinear. By default, Python drops the first category (alphabetically), giving

$$Y = \beta_0 + \beta_{\text{hispanic}} X_{\text{hispanic}} + \beta_{\text{other}} X_{\text{other}} + \beta_{\text{white}} X_{\text{white}} + \epsilon.$$

Question: fit this model. What's the expected earnings disparity between someone who is White and someone who is Hispanic?

Solution: see the companion Colab notebook. The regression coefficients, in terms of conditional expectations, are

$$\begin{aligned} \beta_{\text{white}} &= E[Y \mid X_{\text{white}}{=}1, X_{\text{hispanic}}{=}X_{\text{other}}{=}0] \\ &\quad - E[Y \mid X_{\text{white}}{=}0, X_{\text{hispanic}}{=}X_{\text{other}}{=}0] \\ &= E[Y \mid \text{White}] - E[Y \mid \text{Black}] \approx 4.9, \end{aligned}$$

$$\begin{aligned} \beta_{\text{hispanic}} &= E[Y \mid X_{\text{hispanic}}{=}1, X_{\text{white}}{=}X_{\text{other}}{=}0] \\ &\quad - E[Y \mid X_{\text{hispanic}}{=}0, X_{\text{white}}{=}X_{\text{other}}{=}0] \\ &= E[Y \mid \text{Hispanic}] - E[Y \mid \text{Black}] \approx -0.7. \end{aligned}$$

Our goal is $E[Y \mid \text{White}] - E[Y \mid \text{Hispanic}]$, which we can get by subtracting these two:

$$E[Y \mid \text{White}] - E[Y \mid \text{Hispanic}] = \big(\beta_0 + \beta_{\text{white}}\big) - \big(\beta_0 + \beta_{\text{hispanic}}\big) = \beta_{\text{white}} - \beta_{\text{hispa}}$$

Drill

Counting dummy variables

A predictor records the season a measurement was taken: winter, spring, summer, or fall.

a. How many dummy variables go into the model, and why not one per category?

b. With winter dropped as the baseline, state what β_{summer} means.

c. Using the fitted coefficients, how would you compute $E[Y \mid \text{summer}] - E[Y \mid \text{fall}]$?

Problems

Problem 4.1 — A binary and Normal predictor

Consider the linear regression model

$$Y \mid (X_1, X_2) \sim \text{Normal}(\beta_0 + \beta_1 X_1 + \beta_2 X_2,\ \sigma^2),$$

where the two predictors satisfy $X_1 \sim \text{Bernoulli}(q)$ and $X_2 \mid X_1 \sim \text{Normal}(bX_1, s^2)$.

a. Give at least two real-world examples where this would be a reasonable model relating three variables X_1, X_2, Y.

b. What are $\text{cov}(X_1, X_2)$ and $\text{var}(X_2)$ in terms of the model parameters $q, b, s, \beta_0, \beta_1, \beta_2, \sigma^2$? You may assume $\beta_0 = 0$ for this part and the rest of the exercise — it simplifies the calculations without changing the results.

c. Derive a formula for $\text{cov}(Y, X_1)$ in terms of β_1, q, β_2, b.

d. How does the formula from part (c) relate to the formula for $\text{cov}(Y, X_1)$ in the single-predictor regression model (Unit 3)? For what parameter values do the two formulas coincide? This is a special case of the general relationship between β_1 and covariances that we saw in class.

e. Derive the formula

$$\text{var}(Y) = q(1-q)\left(\beta_1^2 + \beta_2^2 b^2 + 2\beta_1\beta_2 b\right) + \beta_2^2 s^2 + \sigma^2.$$

You'll need the formula for the variance of a sum of two (not necessarily independent) random variables — see the "addition and multiplication" section of the Wikipedia page on variance.

f. The calculation in part (c) lets us answer a classic textbook question, in the more restrictive context of a binary and Normal predictor: is it possible for β_1 and β_2 to *both be negative*, yet the marginal slope of Y vs. X_1 be *positive*? If so, generate simulated data demonstrating this.

Problem 4.2 — Earnings data

Consider the earnings data, loaded with

```
url = (
    "https://raw.githubusercontent.com/avehtari/"
    "ROS-Examples/master/Earnings/data/earnings.csv"
)
df = pd.read_csv(url)
```

As in Unit 3's problems, you'll study the association between earnings and gender, but now with multiple predictors.

a. Perform a linear regression using `statsmodels` with gender and height as predictors.

b. Interpret each regression coefficient (as we did in class for the test-score example).

c. Based on your analysis, which factor — height or gender — is more important?

d. Using the fitted model, predict the chance that someone who is not male and is 5.8ft tall earns more than a male of the same height. To gauge how much height matters, compare this to the chance a male earns more than a non-male, regardless of height.